



LESSON 6-5

Properties of Logarithms

Lesson Overview

Objective

Students will be able to:

- ✓ Use Properties of Logarithms to rewrite logarithmic expressions.
- ✓ Use the Change of Base Formula to evaluate logarithmic expressions and solve equations.

Essential Understanding

Properties of Logarithms can be used to rewrite logarithmic expressions and to evaluate logarithms by changing the base.

Previously in this topic, students:

- Used the inverse relationship between exponential and logarithmic functions to define and graph logarithms.

In this lesson, students:

- Use the inverse relationship between exponential and logarithmic functions to prove Properties of Logarithms.
- Simplify and expand expressions using Properties of Logarithms.

Later in this topic, students will:

- Use Properties of Logarithms to solve exponential and logarithmic equations.

This lesson emphasizes a blend of *conceptual understanding* and *procedural skill and fluency*.

- Students understand the Properties of Logarithms and relate these to the properties of operations and exponents.
- Students evaluate logarithmic expressions.



Glossary is available at SavvasRealize.com.

Vocabulary Builder

REVIEW VOCABULARY

English | Spanish

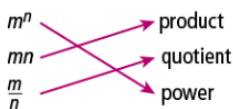
- **logarithm** | *logaritmo*
- **power** | *poder*
- **product** | *producto*
- **quotient** | *cociente*

NEW VOCABULARY

- **Change of Base Formula** | *Fórmula de cambio de base*

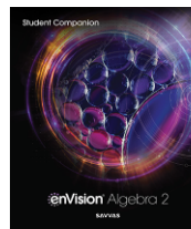
VOCABULARY ACTIVITY

In this lesson students will apply the Properties of Logarithms to rewrite logarithmic expressions. Review the terms *product*, *quotient*, and *power* to preview the Properties of Logarithms. Have students complete this matching activity to choose the term that describes the relationship between m and n in each expression.



Student Companion

Students can do their in-class work for the lesson in their *Student Companion* or on Savvas Realize.



Mathematics Overview

Students apply their understanding of logarithms to prove the Product Property and Quotient Properties of Logarithms. They use these properties to expand logarithmic expressions and write expressions as single logarithms.

Students use the Change of Base Formula to express the solution of an equation as a logarithm.

Applying Math Practices

Reason Abstractly and Quantitatively

Students decontextualize the quantities represented in a logarithmic model to reveal the concentration of hydrogen ions in acid rainfall.

Look For and Make Use of Structure

Students apply general mathematical rules about the properties of logarithms to rewrite logarithmic expressions and solve equations.



Explore & Reason

INSTRUCTIONAL FOCUS Students deduce one of the properties of logarithms when they recognize that the sum of the logarithms of two factors equals the logarithm of their product.

STUDENT COMPANION Students can complete the *Explore & Reason* activity in their *Student Companion*.

Before **WHOLE CLASS**

Implement Tasks that Promote Reasoning and Problem Solving **ETP**

Q: What are some of the features of the graph of $y = \log x$?
[The graph has an x -intercept at 1 and vertical asymptote at the y -axis.]

Q: What kinds of problems is a graph or a table useful in solving?
[Sample: problems that require you to find a pattern]

During **SMALL GROUP**

Support Productive Struggle in Learning Mathematics **ETP**

Q: What do you notice about the labeled points on the graph?
[The x -values 3 and 5 are factors of 15.]

For Early Finishers

Q: Choose four composite numbers. Find the logarithm of the numbers and the logarithms of a pair of their factors. Make a table of your results. Are your results the same as in Part B?
[Yes; the sum of logarithms of the factors is equal to the logarithm of the number.]

After **WHOLE CLASS**

Facilitate Meaningful Mathematical Discourse **ETP**

Ask students to share their answers from Parts B and C. Some students may have trouble extending what they discovered in Part B to Part C.

Q: What do you notice about the values of $\log x$ in the table?
[The sum of the first two values of $\log x$ is equal to the third value of $\log x$.]

Q: How did you apply what you discovered in Part B to Part C?
[Sample: In Part B, I determined that the sum of the logs of two factors is equal to the log of their product. Because 3 and 15 are factors of 45, I predicted $\log 45$ would equal $\log 3 + \log 15$. Similarly, I predicted $\log 75$ would equal $\log 5 + \log 15$.]

Use with EXPLORE & REASON

HABITS OF MIND

Generalize Do you think that the relationships you found in the Explore & Reason activity would also hold for natural logarithms? Give an example.

[yes; $\ln 3 + \ln 5 = \ln 15$]



Critique & Explain is available at SavvasRealize.com.

EXPLORE AND REASON

Look at the graph of $y = \log x$ and the ordered pairs shown.

A. Complete the table.

x	3	5	15
$\log x$			

STUDENT EDITION

6-5 Properties of Logarithms

EXPLORE & REASON

Look at the graph of $y = \log x$ and the ordered pairs shown.

A. Complete the table shown.

x	3	5	15
$\log x$			

B. **Look for Relationships** What is the relationship between the numbers 3, 5, and 15? What is the relationship between the logarithms of 3, 5, and 15?

C. What is your prediction for the value of $\log 45$? $\log 75$? Explain.

How are the properties of logarithms

SAMPLE STUDENT WORK

A.

x	3	5	15
$\log x$	0.477	0.699	1.176

B. $3 \times 5 = 15$, so 3 and 5 are both factors of 15. $0.477 + 0.699 = 1.176$, so $\log 3 + \log 5 = \log 15$.

C. I know that $45 = 3 \times 15$, so $\log 45$ is the same as $\log 3 + \log 15$, meaning the value of $\log 45$ is 1.653. And $75 = 5 \times 15$, so $\log 75 = \log 5 + \log 15$, or about 1.875.



INTRODUCE THE ESSENTIAL QUESTION

Establish Mathematics Goals to Focus Learning **ETP**

Understand that, similar to exponents, there are special properties for products, quotients, and powers relating to logarithms. Students will discover as they work through the examples in the lesson that this is useful when rewriting expressions and solving exponential equations.

Q: What do you notice about the Product Property of Logarithms? [The log of a product equals the sum of the logs of the factors.]

EXAMPLE 1 Prove a Property of Logarithms

Elicit and Use Evidence of Student Thinking **ETP**

Q: Why would the equations $x = \log_b m$ and $y = \log_b n$ be used to start the proof?

[These expressions can be rewritten as exponential equations with a common base. The Product Property of Exponents can then be used to complete the proof.]

Learn Together: Listen Actively When you listen actively, you try to understand what the other person means. Ask What can you do to make sure you hear and understand someone's idea? What can you do or say if you still don't understand?

Examples are available at SavvasRealize.com.

ESSENTIAL QUESTION

How can you use properties of logarithms to simplify expressions and solve logarithmic equations?

CONCEPT

Properties of Logarithms

For positive numbers b , m , and n with $b \neq 1$, the following properties hold.

Product Property of Logarithms $\log_b mn = \log_b m + \log_b n$

Quotient Property of Logarithms $\log_b \frac{m}{n} = \log_b m - \log_b n$

Power Property of Logarithms $\log_b m^p = p \log_b m$

EXAMPLE 1

Prove a Property of Logarithms

How can you prove the Product Property of Logarithms?

Let $x = \log_b m$ and $y = \log_b n$. Then $b^x = m$ and $b^y = n$.

$b^x \cdot b^y = m \cdot n$ Multiply the expressions b^x and b^y .

$b^{x+y} = mn$

$x + y = \log_b mn$ Rewrite the equation in logarithmic form.

$\log_b m + \log_b n = \log_b mn$

Try It! 1. Prove the Quotient Property of Logarithms.

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Try It! Answer

1. Let $x = \log_b m$ and $y = \log_b n$. Then $b^x = m$ and $b^y = n$.

$$\frac{b^x}{b^y} = \frac{m}{n}$$

$$b^{x-y} = \frac{m}{n}$$

$$x - y = \log_b \left(\frac{m}{n} \right)$$

$$\log_b m - \log_b n = \log_b \left(\frac{m}{n} \right)$$

Additional Examples

Go back

ADDITIONAL EXAMPLE 2

ESSENTIAL QUESTION

Use properties of logarithms to expand the expression $\log_5 \left(\frac{3^2}{y^9} \right)$.

CONCEPT

Properties of logarithms

For positive numbers b , m , and n with $b \neq 1$, the following properties hold.

$\log_b mn = \log_b m + \log_b n$ Product Property of Logarithms

$\log_b \frac{m}{n} = \log_b m - \log_b n$ Quotient Property of Logarithms

$\log_b m^p = p \log_b m$ Power Property of Logarithms

$\log_5 \left(\frac{3^2}{y^9} \right) = \log_5 3^2 + \log_5 \frac{1}{y^9} = \log_5 3^2 - \log_5 y^9$

$= \log_5 3^2 - \log_5 y^9$

$= 2 \log_5 3 - 9 \log_5 y$

SOLUTION

Example 2 Students expand logarithmic expressions using the Properties of Logarithms with this additional example.

Q: Which property should you apply first?

[Apply the Quotient Property of Logarithms first to remove the fraction.]

Additional Examples are available at SavvasRealize.com.

Go back

ADDITIONAL EXAMPLE 6

ESSENTIAL QUESTION

Solve $3^x = 10$ using common logarithms.

Then solve $3^x = 10$ using natural logarithms.

Explain the results.

$x = \log_3 10$ $x = \log_3 10$

$x = \frac{\log 10}{\log 3}$ $x = \frac{\ln 10}{\ln 3}$

$x \approx 2.096$ $x \approx 2.096$

Check: $3^{2.096} \approx 10$

You can solve an exponential equation by using common logarithms or by using natural logarithms because both methods give the same answer.

SOLUTION

Example 6 Students compare the results of solving an exponential equation using the Change of Base Formula with different bases.

Q: What would the solution be if you used base 2?

[The solution would be the same.]



EXAMPLE 2 Expand Logarithmic Expressions

Facilitate Meaningful Mathematical Discourse ETP

Q: Why might it be useful to know how to expand logarithmic expressions?

[It may be easier to solve logarithmic equations when each side is written in expanded form.]

Try It! Answers

2. a. $3 \log_7 r + 4 \log_7 t - \log_7 v$
b. $\ln 7 - 2 \ln 3 - 2 \ln 5$

EXAMPLE 3 Write Expressions as Single Logarithms

Pose Purposeful Questions ETP

Q: How do the processes of expanding a logarithm and writing an expression as a single logarithm relate to each other?

[The process of writing expressions as single logarithms is the reverse of the process of expanding logarithmic expressions.]

Try It! Answers

3. a. $\log_2 \left(\frac{c^5}{n^7} \right)$
b. $\ln 98$

Elicit and Use Evidence of Student Thinking ETP

Q: Is it necessary to apply the Property of Logarithms in a specific order? [Sample: Yes, you cannot apply the Quotient Property until after you apply the Power Property.]

Use with EXAMPLES 2 & 3

HABITS OF MIND

Make Sense and Persevere Using the fact that $\log 2 \approx 0.3010$ and $\log 3 \approx 0.4771$, what is $\log 18$? Explain.

[approx. 1.2553; since $18 = 2 \times 3 \times 3$,
 $\log 18 = \log 2 + \log 3 + \log 3$]



Examples are available at SavvasRealize.com.

EXAMPLE 2 Expand Logarithmic Expressions

How can you use the properties of logarithms to expand each expression?

A. $\log_5(a^2b^7)$
 $\log_5(a^2b^7) = \log_5(a^2) + \log_5(b^7)$
 $= 2 \log_5 a + 7 \log_5 b$

B. $\ln \left(\frac{25}{3} \right)$
 $\ln \left(\frac{25}{3} \right) = \ln(25) - \ln(3)$
 $= \ln(5^2) - \ln 3$
 $= 2 \ln 5 - \ln 3$

Try It! 2. Use the properties of logarithms to expand each expression.

a. $\log_7 \left(\frac{r^3 s^4}{v} \right)$ b. $\ln \left(\frac{7}{25} \right)$

EXAMPLE 3 Write Expressions as Single Logarithms

What is each expression written as a single logarithm?

A. $4 \log_4 m + 3 \log_4 n - \log_4 p$
 $4 \log_4 m + 3 \log_4 n - \log_4 p$
 $= \log_4(m^4) + \log_4(n^3) - \log_4 p$
 $= \log_4(m^4 n^3) - \log_4 p$
 $= \log_4 \left(\frac{m^4 n^3}{p} \right)$

B. $3 \ln 2 - 2 \ln 5$
 $3 \ln 2 - 2 \ln 5 = \ln(2^3) - \ln(5^2)$
 $= \ln \left(\frac{2^3}{5^2} \right)$
 $= \ln \left(\frac{8}{25} \right)$

Try It! 3. Write each expression as a single logarithm.

a. $5 \log_2 c - 7 \log_2 n$ b. $2 \ln 7 + \ln 2$

MAKE SENSE AND PERSEVERE
Expanding logarithmic expressions requires the use of a variety of properties. Look at the whole expression first to determine the sequence of properties that will be used.

STUDY TIP
Recall that the Properties of Logarithms are each associated with a different operation. Addition signals the Product Property, subtraction signals the Quotient Property, and multiplication by a constant signals the Power Property.

ELL English Language Learners (Use with EXAMPLE 3)

SPEAKING BEGINNING In small groups, ask students to discuss how many different meanings of the word *log* they can think of. Encourage them to think of how *log* can be used as a noun and how it can be used as a verb.

Q: What are some meanings for the word *log* your group came up with?
 [part of a tree trunk, a recording of data or events, a device that determines the speed of a ship, to cut trees down, to keep a record of, to make a certain speed or travel for a certain amount of time]

Q: Why do we call a logarithm a *log* when we mean something completely different?
 [Log is like a nickname for logarithm.]

LISTENING DEVELOPING Review the example and the study tip. Draw students' attention to the words *single* and *signal*. Provide sentence frames for students to complete with either *single* or *signal* (i.e. A red light _____ a car to stop. He was so quiet he didn't say a _____ word.).

Q: Do signals give information?
 [Yes; they tell us when to stop and to go.]

Q: Is the word *multiple* similar to the word *single*?
 [No; they are antonyms.]

WRITING EXPANDING After reviewing the example and the study tip, have students spend 2 to 5 minutes writing about how they could use the signals in the problem to write single logarithms.

Q: Why are the expressions at the beginning of the examples not considered to be single?
 [They are expressions with more than one term.]

Q: What signals should you look for first when simplifying an expression to a single logarithm?
 [multiplication by a constant to signal the Power Property]



STEP 2 Understand & Apply

EXAMPLE 4 Apply Properties of Logarithms

Facilitate Meaningful Mathematical Discourse ETP

Q: What does the + in H^+ represent? Does it affect how you solve the equation?

$[H^+]$ is the concentration of hydrogen ions in a solution. The + indicates that the hydrogen ions have a positive charge. H^+ is a variable, so the + does not affect how the equation is solved.]

Try It! Answer

4. 10^{-3} moles per liter, or 0.001 moles per liter

HABITS OF MIND

Generalize What types of numbers have logarithms that are negative? Explain.

[Real numbers between 0 and 1; a positive number raised to a negative exponent is equal to a number between 0 and 1.]

Use with EXAMPLE 4

EXAMPLE 5 Evaluate Logarithmic Expressions by Changing the Base

Pose Purposeful Questions ETP

Q: What does each variable represent when you use the Power Property of Logarithms?

$[n = \log_2 3, b = 10, \text{ and } m = 2]$

Try It! Answers

5. a. 2.807

b. 0.683

Elicit and Use Evidence of Student Thinking ETP

Q: Do you need to go through all of the steps shown in the example to complete the Try It? Explain.

[No; The example illustrated how the Change of Base Formula was derived. To complete the Try It, the formula can be used directly.]

Examples are available at SavvasRealize.com.

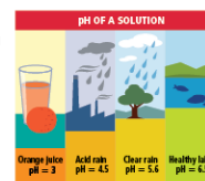
APPLICATION

EXAMPLE 4 Apply Properties of Logarithms

The pH of a solution is a measure of its concentration of hydrogen ions. This concentration (measured in moles per liter) is written $[H^+]$ and is given by the formula

$$pH = \log_{10} \frac{1}{[H^+]}$$

What is the concentration of hydrogen ions in the acid rainfall?



Solve for $[H^+]$

$$4.5 = \log_{10} \frac{1}{[H^+]}$$

$$4.5 = \log_{10} 1 - \log_{10} [H^+]$$

$$-4.5 = \log_{10} [H^+]$$

$$10^{-4.5} = [H^+]$$

The concentration of hydrogen ions in the acid rainfall is $10^{-4.5} \approx 0.0000316$ moles per liter.

Try It! 4. What is the concentration of hydrogen ions in a liter of orange juice?

CONCEPTUAL UNDERSTANDING

EXAMPLE 5 Evaluate Logarithmic Expressions by Changing the Base

How can you use base 10 logarithms to evaluate base 2 logarithms?

To evaluate $\log_2 3$ with a calculator, you need to express $\log_2 3$ in terms of base 10 logarithms.

$$\log_2 3 = \frac{\log_2 3 \cdot \log_2 2}{\log_2 2}$$

$$= \frac{\log_{10} 3 \cdot \log_{10} 2}{\log_{10} 2}$$

$$= \frac{\log_{10} 3}{\log_{10} 2}$$

$$= \frac{0.477}{0.301} \approx 1.585$$

This illustrates the **Change of Base Formula**:

$$\text{For positive numbers } m, b, \text{ and } a, \text{ with } b \neq 1 \text{ and } a \neq 1, \log_b m = \frac{\log_a m}{\log_a b}.$$

Try It! 5. Estimate the value of each logarithm. Then use a calculator to find the value of each logarithm to the nearest thousandth.
a. $\log_2 7$ b. $\log_3 3$

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RII Extend Student Thinking

USE WITH EXAMPLE 3 Have students explore the effect of parentheses on writing expressions as single logarithms.

- Give students these examples.

$$4 \log x + 3 \log 2 - (2 \log y + \log 5)$$

Q: Write this expression as a single logarithm.

$$\left[\log \left(\frac{8x^4}{5y^2} \right) \right]$$

Q: Write $4 \log x + 3 \log 2 - 2 \log y + \log 5$ as a single expression.

$$\left[\log \left(\frac{40x^4}{y^2} \right) \right]$$

Q: How did removing the parentheses from

$4 \log x + 3 \log 2 - (2 \log y + \log 5)$ affect writing the expression as a single logarithm?

[Removing the parentheses changed the expression to have only one factor in the denominator.]

RII Support Struggling Students

USE WITH EXAMPLE 5 Some students have difficulty understanding the Change of Base Formula.

- Have students analyze each step of the process of deriving the Change of Base Formula.

Q: What is the purpose of multiplying by 1 in the form of $\frac{\log_2 2}{\log_2 2}$? [Most calculators cannot evaluate logs with a base of 2, so multiplying by $\frac{\log_2 2}{\log_2 2}$ is the first step in rewriting the expression with a base of 10.]

Q: Explain how to move from the second step to the third step. [Exponents and logarithms are inverse operations, so in the expression $\log_2 \log_2 3$, the exponent of $\log_2$ undoes the 2, leaving only $\log 3$.]

Q: Why can you use a calculator to evaluate $\frac{\log 3}{\log 2}$? [Both logarithms have a base of 10, which a calculator can evaluate.]



EXAMPLE 6 Use the Change of Base Formula

Implement Tasks that Promote Reasoning and Problem Solving ETP

Q: Why can you solve the equation using both base 10 logarithms and natural logarithms?

[According to the Change of Base Formula, the only restrictions on the base are that it is positive and it cannot equal 1. So logarithms of base 10, e , or any other base can be used.]

Try It! Answer

6. $\left(\frac{\log 15}{\log 3}\right)$; 2.465

Common Error

Try It! 6 Students may divide each side by the base of the exponent instead of using logarithms. Emphasize that the equation is an exponential equation and that logarithms are the inverse of exponents.

Use with EXAMPLES 5 & 6

HABITS OF MIND

Use Appropriate Tools Why is the Change of Base Formula useful when evaluating a logarithm with a calculator?

[Standard calculators do not have an option to take the logarithm of something with a base other than 10 or e , so the Change of Base Formula gives you a way to solve the expression using a calculator.]

CONCEPT SUMMARY Properties of Logarithms

Q: How are the properties of logarithms useful when solving problems?

[The properties of logarithms help you rewrite logarithms into forms that are more useful to solve specific problems. The Change of Base formula can be used to estimate the value of a logarithm.]

Do You UNDERSTAND? | Do You KNOW HOW?

Common Error

Exercise 5 Students may incorrectly use the product property to write the expression as $\ln(30st)$ instead of $\ln(s^5t^6)$. Explain that the first step is to rewrite $5 \ln s + 6 \ln t$ as $\ln s^5 + \ln t^6$ using the product property.



Concept Summary, Do You Understand, and Do You Know How are available at SavvasRealize.com.

EXAMPLE 6 Use the Change of Base Formula

What is the solution of the equation $2^x = 7$? Express the solution as a logarithm and then evaluate. Round to the nearest thousandth.

$$\begin{aligned} 2^x &= 7 \\ x &= \log_2 7 \\ x &= \frac{\log 7}{\log 2} \\ x &\approx 2.807 \end{aligned}$$

Check $2^{2.807} \approx 7$

STUDY TIP

The equation $2^x = 7$ can also be solved using natural logarithms.

Try It! 6 What is the solution to the equation $3^x = 15$? Express the solution as a logarithm, make an estimate, and then evaluate. Round to the nearest thousandth.

CONCEPT SUMMARY Properties of Logarithms

	Product Property	Quotient Property	Power Property	Change of Base
ALGEBRA	$\log_b(mn) = \log_b m + \log_b n$	$\log_b\left(\frac{m}{n}\right) = \log_b m - \log_b n$	$\log_b(m^n) = n \cdot \log_b m$	$\log_b m = \frac{\log_a m}{\log_a b}$
WORDS	The log of a product is the sum of the logs.	The log of a quotient is the difference of the logs.	The log of a number raised to a power is the power multiplied by the log of the number.	The log base b of a number is equal to the log base a of the number divided by the log base a of b .
NUMBERS	$\log_2(20) = \log_2(4) + \log_2(5)$	$\log_{10}\left(\frac{2}{3}\right) = \log_{10} 2 - \log_{10} 3$	$\log_3(16) = 4 \cdot \log_3 2$	$\log_5 7 = \frac{\log 7}{\log 5}$

Do You UNDERSTAND?

- ESSENTIAL QUESTION** How are the properties of logarithms used to simplify expressions and solve logarithmic equations?
- Vocabulary** While it is not necessary to change to base 10 when applying the Change of Base Formula, why is it common to do so?
- Error Analysis** Amanda claimed the expanded form of the expression $\log_4(c^2d^5)$ is $5 \log_4 c + 5 \log_4 d$. Explain the error Amanda made.

Do You KNOW HOW?

- Use the properties of logarithms to expand the expression $\log_6\left(\frac{49}{5}\right)$.
- Use the properties of logarithms to write the expression $5 \ln s + 6 \ln t$ as a single logarithm.
- Use the formula $\text{pH} = \log_{10} \frac{1}{[\text{H}^+]}$ to write an expression for the concentration of hydrogen ions, $[\text{H}^+]$, in a container of baking soda with a pH of 8.9.

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Answers

- When multiple logarithms with the same base are added or subtracted, you can combine them into a single expression involving the logarithm of a product or quotient. Equations involving these simpler expressions are often easier to solve.
- This makes it easier to find the value of the expression on the calculator.
- Amanda used the wrong exponent for c . The expression should be $2 \log_4 c + 5 \log_4 d$.
- $2 \log_6 7 - \log_6 5$
- $\ln(s^5t^6)$
- $10^{-8.9}$ or 1.259×10^{-9}

STEP 3 Practice & Problem Solving



PRACTICE & PROBLEM SOLVING



Practice & Problem Solving and video tutorials are available at SavvasRealize.com.

Scan the QR code to access a video tutorial.

SavvasRealize.com



Lesson Practice You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: Lesson Practice

Adaptive Practice

You may also take advantage of the bank of exercises for assigning additional practice.

Assignment Guide

On-Level	Advanced
7–27, 32–40	7–15, 20–40

Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways.

For example:

Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.

Understand, Apply.....2 points each
Practice.....1 point each
Assessment Practice.....1 point each
Performance Task.....3 points

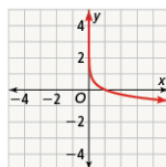
Item Analysis

Example	Items	DOK
1	13	3
2	14–17	1
	7, 9	2
3	18–22, 39	1
	8, 11, 37	2
4	23, 36, 40	2
5	24–29, 38	1
	12	2
6	30–35	1
	10	2

PRACTICE & PROBLEM SOLVING

UNDERSTAND

- Use Structure** Without applying the Change of Base Formula, explain how to use $\log_3 2 \approx 0.631$ and $\log_3 5 \approx 1.465$ to approximate $\log_3 \left(\frac{2}{5}\right)$.
- Communicate Precisely** Explain what is meant by expanding a logarithmic expression. How are the processes of expanding logarithmic expressions and writing logarithmic expressions as a single logarithm related?
- Higher Order Thinking** The graph of $y = \log\left(\frac{1}{x}\right)$ and $y = -\log x$ are shown. Notice the graph is the same for both equations. Use properties of logarithms to explain why the graphs are the same.



- Communicate Precisely** Saachi used the Change of Base Formula to solve the equation $6^x = 72$ and found that $x = 2.387$. How can Saachi check her solution?
- Error Analysis** Describe and correct the error a student made in writing the logarithmic expression in terms of a single logarithm.

$$\log_3 2 + \frac{1}{2} \log_3 5 = \log_3 2y^2$$



- Error Analysis** A student wants to approximate $\log_2 9$ with her calculator. She enters the equivalent expression $\frac{\ln 2}{\ln 9}$, but the decimal value is not close to her estimate of 3. What happened?

$$\log_2 9 = \frac{\ln 2}{\ln 9}$$



PRACTICE

- Use the properties of exponents to prove the Power Property of Logarithms. SEE EXAMPLE 1

Use the properties of logarithms to expand each expression. SEE EXAMPLE 2

$$14. \log_5 \left(\frac{2}{3}\right)$$

$$15. \log_6 (2m^5n^3)$$

$$16. \ln 2x^5$$

$$17. \log_2 \left(\frac{x}{5y}\right)$$

Use the properties of logarithms to write each expression as a single logarithm. SEE EXAMPLE 3

$$18. 9 \ln x - 6 \ln y \quad \ln \left(\frac{x^9}{y^6}\right) \quad 19. \log_5 6 + \frac{1}{2} \log_5 y \log_5 (6y^2)$$

$$20. 2 \log 10 + 4 \log (3x) \quad 21. \frac{1}{3} \ln 27 - 3 \ln (2y)$$

$$22. 8 \log_3 2 + 5 \log_3 c + 7 \log_3 d$$

$$23. \text{Use properties of logarithms to show that } \text{pH} = \log_{10} \frac{1}{[H^+]} \text{ can be written as } \text{pH} = -\log [H^+]. \text{ SEE EXAMPLE 4}$$

Use the Change of Base Formula to evaluate each logarithm. Round to the nearest thousandth.

SEE EXAMPLE 5

$$24. \log_4 9 \quad 1.585$$

$$25. \log_6 5 \quad 0.898$$

$$26. \ln 3 \quad 1.099$$

$$27. \log_2 7 \quad 2.807$$

$$28. \log_9 12 \quad 1.131$$

$$29. \ln 23 \quad 3.135$$

Use the Change of Base Formula to solve each equation for x . Give an exact solution as a logarithm and an approximate solution rounded to the nearest thousandth. SEE EXAMPLE 6

$$30. 3^x = 4 \left(\frac{\log 4}{\log 3}\right); 1.262 \quad 31. 5^x = 11 \left(\frac{\log 11}{\log 5}\right); 1.490$$

$$32. 8^x = 10 \left(\frac{\log 10}{\log 8}\right); 1.107 \quad 33. 2^x = 30 \left(\frac{\log 30}{\log 2}\right); 4.907$$

$$34. 7^x = 100 \left(\frac{\log 100}{\log 7}\right); 2.367 \quad 35. 4^x = 55 \left(\frac{\log 55}{\log 4}\right); 2.891$$

LESSON 6-5 Properties of Logarithms 319

Answers

- Use the Quotient Property to rewrite $\log_3 \left(\frac{2}{5}\right)$ as $\log_3 2 - \log_3 5$. Substitute 0.631 for $\log_3 2$ and 1.465 for $\log_3 5$, then subtract to approximate the logarithm as -0.834 .
- To expand, separate the factors inside the logarithm into separate sums of logarithms. To write a single logarithmic expression, combine sums of logarithms into a single logarithm of the product.

- $y = \log \left(\frac{1}{x}\right) = \log 1 - \log x = 0 - \log x = -\log x$
- Saachi can check if $6^{2.387} = 72$.
- The student used 2 as an exponent instead of $\frac{1}{2}$. The correct expression is $\log_3 2y^{\frac{1}{2}}$.

See answers for 12–17, 23 on the next page.



Practice

Answers

12. The student applied the change of base property incorrectly.

13. Let $\log_b(m^n) = x$.

$$\begin{aligned} b^x &= m^n \\ (b^x)^{\frac{1}{n}} &= (m^n)^{\frac{1}{n}} \\ b^{\frac{x}{n}} &= m \\ \frac{x}{n} &= \log_b m \\ x &= n \log_b m \\ \log_b(m^n) &= n \log_b(m) \end{aligned}$$

14. $\log_5 2 - \log_5 3$

15. $\log_6 2 + 5 \log_6 m + 3 \log_6 n$

16. $\ln 2 + 5 \ln x$

17. $\log_2 x - \log_2 5 - \log_2 y$

23. $\text{pH} = \log\left(\frac{1}{[\text{H}^+]}\right) = \log 1 - \log [\text{H}^+] = 0 - \log [\text{H}^+] = -\log [\text{H}^+]$

36. a. 80 dB

b. 10^{-8} W/m^2

c. 10,000 times

37. a. $A = \log(5t^2)$

b. 4,325 people

40. Part A 2.3

Part B $10^{0.9}$, or about 7.94, times greater

Part C The magnitude of the more intense earthquake is about 2.2 greater than the magnitude of the less intense earthquake.

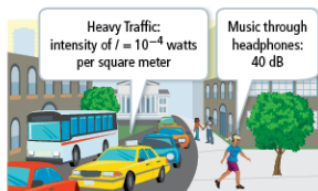


Practice & Problem Solving is available at SavvasRealize.com.

PRACTICE & PROBLEM SOLVING

APPLY

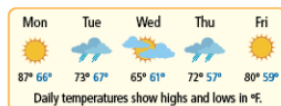
36. **Make Sense and Persevere** The loudness of sound is measured in decibels. For a sound with intensity I (in watts per square meter), its loudness $L(I)$ (in decibels) is modeled by the function $L(I) = 10 \log_{10} \frac{I}{I_0}$, where I_0 represents the intensity of a barely audible sound (approximately 10^{-12} watts per square meter).



- Find the decibel level of the sound made by the heavy traffic.
- Find the intensity of the sound that is made by music playing at 40 decibels.
- How many times as great is the intensity of the traffic than the intensity of the music?

37. **Model with Mathematics** Miguel collected data on the attendance at an amusement park and the daily high temperature. He found that the model $A = 2 \log t + \log 5$ approximated the attendance A , in thousands of people, at the amusement park, when the daily high temperature is t degrees Fahrenheit.

- Use properties of logarithms to simplify Miguel's formula.
- The daily high temperatures for the week are below.



What is the expected attendance on Wednesday? Round to the nearest person.

ASSESSMENT PRACTICE

38. Select all the expressions equal to $\log_2 3$.

- ☒ $\frac{\log 3}{\log 2}$
- ☐ $\frac{\log 2}{\log 3}$
- ☒ $\frac{\log_2 3}{\log_2 2}$
- ☐ $\frac{\log_2 2}{\log_2 3}$
- ☐ $\frac{\log_2 3}{\log_2 3}$

39. **SAT/ACT** Use the properties of logarithms to write the following expression in terms of a single logarithm.

$$2(\log_3 20 - \log_3 4) + 0.5 \log_3 4$$

- ☐ A $\log_3 4$
- ☐ B $\log_3 5$
- ☐ C $\log_3 25$
- ☒ D $\log_3 50$

40. **Performance Task** The magnitude, or intensity, of an earthquake is measured on the Richter scale. For an earthquake where the amplitude of its seismographic trace is A , its magnitude is modeled by the function:

$$R(A) = \log \frac{A}{A_0}$$

where A_0 represents the amplitude of the smallest detectable earthquake.

Part A An earthquake occurs with an amplitude 200 times greater than the amplitude of the smallest detectable earthquake, A_0 . What is the magnitude of this earthquake on the Richter scale?

Part B Approximately how many times as great is the amplitude of an earthquake measuring 6.8 on the Richter scale than the amplitude of an earthquake measuring 5.9 on the Richter scale?

Part C Suppose the intensity of one earthquake is 150 times as great as that of another. How much greater is the magnitude of the more intense earthquake than the less intense earthquake?



STEP 4 Assess & Differentiate

LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.

Students can take the Lesson Quiz online or you can download a printable copy from [SavvasRealize.com](https://www.savvasrealize.com). The Lesson Quiz is also available in the *Assessment Resources* book.

Item Analysis

Item	DOK	Skills Review and Practice
1	3	F71
2	2	F71
3	2	F71
4	2	F71
5	2	F71



Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

I Intervention	0–3 points	• Reteach to Build Understanding • Mathematical Literacy and Vocabulary • Lesson Virtual Nerd videos
O On-Level	4 points	• Enrichment
A Advanced	5 points	• Enrichment



Lesson Quiz is available at [SavvasRealize.com](https://www.savvasrealize.com).

ASSESSMENT RESOURCES

Name _____

enVision Algebra 2

SavvasRealize.com

6-5 Lesson Quiz

Properties of Logarithms

1. Select the steps to write the expression $5\log_3 a + \log_3 b - 2\log_3 c$ as a single logarithm using the properties of logarithms.

☐ Power Property

☐ Quotient Property

☐ Product Property

$$5\log_3 a + \log_3 b - 2\log_3 c = \log_3 a^5 + \log_3 b - \log_3 c^2$$

☒ Power Property

$$= \log_3 a^5 b - \log_3 c^2$$

☒ Product Property

$$= \log_3 \left(\frac{a^5 b}{c^2} \right)$$

☒ Quotient Property

2. The pH of a solution is a measure of its concentration of hydrogen ions. This concentration, written $[H^+]$ and measured in moles per liter, is given by the formula $pH = \log_{10} \frac{1}{[H^+]}$. What is the concentration of hydrogen ions in a liter of vinegar that has a pH level of 2.5?

☐ A. $10^{2.5}$

☒ B. $10^{-2.5}$

☐ C. $\log(2.5)$

☐ D. $\log(-2.5)$

3. Use the properties of logarithms to write the expression $\frac{1}{2}\ln 4 + 2\ln x$ as a single logarithm.

☐ A. $4\ln(2x)$

☒ B. $\ln(2x^2)$

☐ C. $\ln(4x)$

☐ D. $\ln(4x^2)$

4. Select all the logarithmic expressions that have been evaluated correctly.

☐ A. $\log_3 8 = 0.43$

☐ D. $\log_2 32 = 1.51$

☒ B. $\log_3 6 = 1.63$

☐ E. $\log_4 7 = 2.21$

☒ C. $\log_4 5 = 1.16$

5. Solve the equation $5^x = 7$ for x . Select the exact solution as a logarithm and use the change of base formula to find an approximate solution.

☐ A. $x = \log_7 5$

☐ E. $x = \log^5$

☒ B. $x = \log_5 7$

☐ F. $x = \log^7$

☐ C. $x = 0.827$

☐ G. $x = 0.699$

☒ D. $x = 1.209$

☐ H. $x = 1.845$

enVision® Algebra 2 • Assessment Sourcebook



DIFFERENTIATED RESOURCES

I = Intervention **O** = On-Level **A** = Advanced

Reteach to Build Understanding **I**

Provides scaffolded reteaching for the key lesson concepts.

MathXL® for School

Name _____

6-5 Reteach to Build Understanding

Properties of Logarithms

1. Complete the table below by filling in the blanks.

Definition of Logarithm	Logarithm	Exponent
$y = \log_b x$ if only $b^{y^x} = x$	$x = \log_b y$ if only $b^{x^y} = y$	
Product Property	$\log_b(mn) = \log_b m + \log_b n$	$\log_b 18 = \log_b(2) + \log_b(9)$
Quotient Property	$\log_b \frac{m}{n} = \log_b m - \log_b n$	$\log_b 18 = \log_b(6) - \log_b(1/3)$
Power Property	$\log_b(m^p) = p \log_b m$	$\log_b(27) = 3 \log_b(3)$
Change of Base Property	$\log_b m = \frac{\log_k m}{\log_k b}$	$\log_2 8 = \frac{\log 8}{\log 2}$

2. Describe and correct the error a student made in writing $\log_2 2 + 2 \log_2 x$ in terms of a single logarithm.

$\log_2 2 + 2 \log_2 x$
 $= \log_2 2 + \log_2 x^2$ Use Power Property of Logarithms.
 $= \log_2 (2 \cdot x^2) = 2$ Use Product Property of Logarithms.
 Student used the Product Property of Logarithms incorrectly by adding instead of multiplying. Since $\log_b(mn) = \log_b m + \log_b n$, $\log_2 2 + \log_2 x^2 = \log_2 (2 \cdot x^2)$.

3. Use the properties of logarithms to complete the following expressions.

a. Product Property: $\log_2 (a^3 b^4) = \log_2 (a^3) + \log_2 (b^4)$
 b. Power Property: $\log_2 (a^3) = \log_2 (b^4) = 3 \log_2 a + 4 \log_2 b$
 c. Change of Base: $\log_2 4 = \frac{\log 4}{\log 2}$
 d. Quotient Property: $\log_2 \left(\frac{1}{3}\right) = \log_2 3 - \log_2 4$

Additional Practice **I O**

Provides extra practice for each lesson.

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6-5 Additional Practice

Properties of Logarithms

Use the properties of logarithms to expand each expression.

1. $\ln(a^2 b^3)$ 2. $\ln(x^4)$ 3. $\log_3(a^2 b^3 c)$
 $4 \ln a + 7 \ln b$ $4 \ln x$ $2 \log_3 a + 3 \log_3 b + \log_3 c$

4. $\log\left(\frac{1}{x}\right)$ 5. $\log_5\left(\frac{1}{x}\right)$ 6. $\log\left(\frac{1}{x}\right)$
 $x \log 7 - x \log 8$ $\log_5 x - (\log_5 7 + \log_5 y)$ $\log a - 2 \log b$

Use the properties of logarithms to write each expression as a single logarithm.

7. $3 \log 4 - 2 \log 7$ 8. $2 \ln 4 + 2 \ln 5$ 9. $2 \log_2 a + 5 \log_2 b$
 $\log\left(\frac{64}{7^2}\right) = \log\left(\frac{64}{49}\right)$ $\ln 400$ $\log_4(a^2 b^5)$

10. $\log 4 + \log 5 = \log 7$ 11. $2 \log 2 + 5 \log(2x)$ 12. $4 \log_2 a - 7 \log_2 b$
 $\log 140$ $\log(4 \times (2x)^5)$ $\log_6\left(\frac{a^4}{b^7}\right)$

13. Use the formula $\text{pH} = \log\left(\frac{1}{[\text{H}^+]}\right)$ to write an expression for the concentration of hydrogen ions in a liter of a sports drink that has a pH level of 2.5. What is the concentration of hydrogen ions? $2.5 = \log\left(\frac{1}{[\text{H}^+]}\right)$, $[\text{H}^+] = 10^{-2.5}$

The concentration of hydrogen ions in a liter of a sports drink is $10^{-2.5} \approx 0.003162$ mole.

Use the Change of Base Formula to evaluate each logarithm. Round to the nearest thousandth, if necessary.

14. $\log_2 10 \approx 3.322$ 15. $\log_2 7 \approx 2.807$ 16. $\log_2 13 \approx 3.700$
 17. $\log_2 9 \approx 3.170$ 18. $\ln e \approx 1.594$ 19. $\log_2 9 \approx 3.170$

Use the Change of Base Formula to solve each equation for x . Give an exact solution as a logarithm and an approximate solution rounded to the nearest thousandth.

20. $5^x = 7$ $\log_5 7 = \log 7$ $\log_5 20 = \log 20$ $22. 2^x = 42$ $\log_2 42 = \log 42$
 $\log_5 7 = \frac{\log 7}{\log 5}$ $\log_5 20 = \frac{\log 20}{\log 5}$ $\log_2 42 = \frac{\log 42}{\log 2}$
 ≈ 1.209 ≈ 2.161 ≈ 1.921

23. $4^x = 77$ $\log_4 77 = \log 77$ $\log_4 50 = \log 50$ $24. 9^x = 50$ $\log_9 50 = \log 50$
 $\log_4 77 = \frac{\log 77}{\log 4}$ $\log_4 50 = \frac{\log 50}{\log 4}$ $\log_9 50 = \frac{\log 50}{\log 9}$
 ≈ 3.133 ≈ 3.133 ≈ 1.881

26. Explain why $\frac{2}{3} \neq \log_3 \frac{2}{3}$.
 $\frac{2}{3} = 0.667$; $\log_3 \frac{2}{3} = 0.631$; Since $0.667 \neq 0.631$, therefore $\frac{2}{3} \neq \log_3 \frac{2}{3}$.

Enrichment **O A**

Presents engaging problems and activities that extend the lesson concepts.

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6-5 Enrichment

Properties of Logarithms

Predict whether each pair of expressions will represent the same graph. Next, check by using your calculator to graph each pair of expressions. Then, use the properties of logarithms to explain why the graphs are the same or different.

1. $y = \ln\left(\frac{1}{x}\right)$ and $y = \ln(x)$ No; using the Quotient Property, $y = \ln\left(\frac{1}{x}\right) = -\ln x$, which is not equivalent to $y = \ln(x)$.
 2. $y = \log_2 45$ and $y = \log_2 5 + \log_2 9$ Yes; using the Product Property, $y = \log_2 45 = \log_2 5 + \log_2 9$.
 3. $y = \log(x)$ and $y = 3 \log x$ No; using the Product Property, $y = \log(3x) = \log 3 + \log x$ which is not equivalent to $y = 3 \log x$.
 4. $y = \log(x^2)$ and $y = 2 \log(x)$ Yes; using the Power Property, $y = \log(x^2) = 2 \log(x)$.

Fill in the blank with the letter of the graph that matches the given function. Some letters may be used more than once.

5. $y = \log\left(\frac{1}{x}\right)$ **A**
 6. $y = \log 2x$ **C**
 7. $y = \log x$ **A**
 8. $y = \log(x^2)$ ($x > 0$) **B**
 9. $y = 2 \log x$ **B**

A. B. C.

Mathematical Literacy and Vocabulary **I O**

Helps students develop and reinforce understanding of key terms and concepts.

Name _____

6-5 Mathematical Literacy and Vocabulary

Properties of Logarithms

Product Property	Quotient Property	Power Property	Change of Base
$\log_b(mn) = \log_b m + \log_b n$	$\log_b\left(\frac{m}{n}\right) = \log_b m - \log_b n$	$\log_b(m^p) = p \log_b m$	$\log_b m = \frac{\log_k m}{\log_k b}$
Example: $\log_2(18) = \log_2(2) + \log_2(9)$	Example: $\log_2\left(\frac{1}{3}\right) = \log_2(1) - \log_2(3)$	Example: $\log_2(27) = 3 \log_2(3)$	Example: $\log_2 8 = \frac{\log 8}{\log 2}$

Identify the property that is demonstrated by each equation.

1. $\log_2 19 = \frac{\log 19}{\log 2}$ **Change of Base**
 2. $\log_2\left(\frac{1}{3}\right) = \log_2(1) - \log_2(3)$ **Quotient Property**
 3. $\log_{10}(27) = 3 \log_{10}(3)$ **Power Property**
 4. $\log_2(27) = \log_2(3) + 2 \log_2(3)$ **Product Property**
 5. $\log_2(77) = \log_2(7) + \log_2(11)$ **Product Property**

Use the properties of logarithms to complete each expression.

6. $\log_{10}(8) + \log_{10}(5) = \log_{10}(40)$
 7. $\log \dots 7$ $36 = 2 \log_2 6$
 8. $\log_2(7) = \frac{\log 7}{\log 2}$

Use the properties of logarithms to write each expression as a single logarithm.

9. $\log_{12} 77 - \log_{12} 7 = \log_{12}(11)$
 10. $\ln(5) + \ln(8) = \ln(40)$
 11. $3 \log_4(2) + \log_4(7) + \log_4(8) = \log_4(336)$

Digital Differentiated Resources

The Reteach to Build Understanding, Additional Practice, and Enrichment activities are available as digital assignments. These activities are automatically assigned when students complete the lesson quiz online and are automatically scored.

